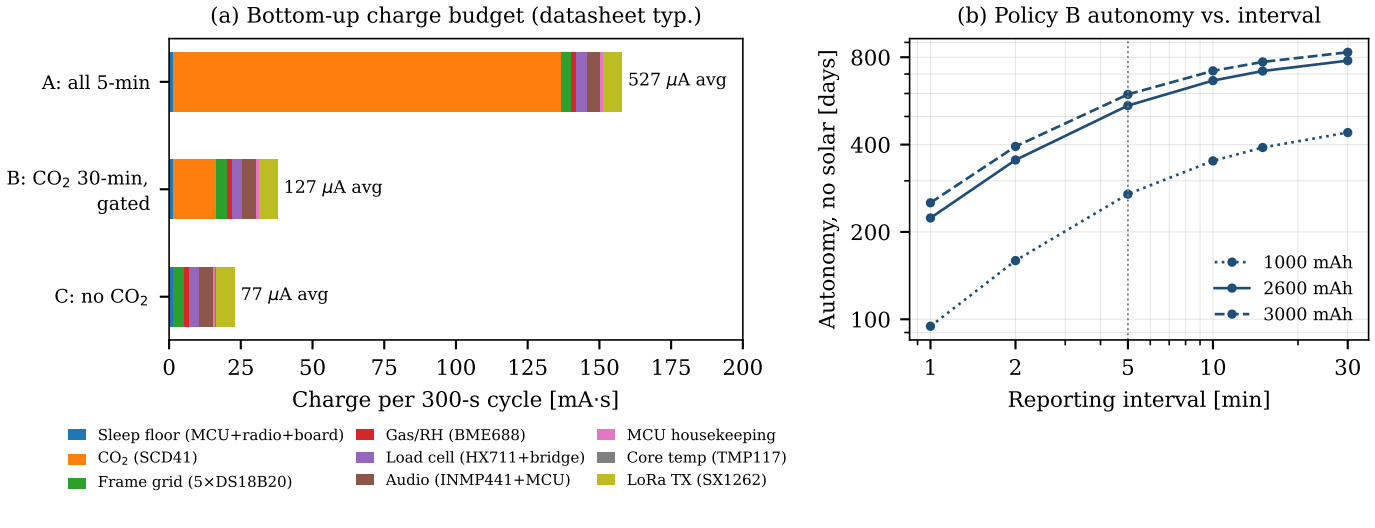
(a) Bottom-up charge budget (datasheet typ.)
A: all 5-min
B: CO2 30-min, gated
C: no CO2
527 μA avg
127 μA avg
77 μA avg
0
25
50
75
100
125
150
175
200
Charge per 300-s cycle [mA·s]
Sleep floor (MCU+radio+board)
CO2 (SCD41)
Frame grid (5×DS18B20)
Gas/RH (BME688)
Load cell (HX711+bridge)
Audio (INMP441+MCU)
MCU housekeeping
Core temp (TMP117)
LoRa TX (SX1262)
(b) Policy B autonomy vs. interval
Autonomy, no solar [days]
800
400
200
100
1
2
5
10
30
Reporting interval [min]
1000 mAh
2600 mAh
3000 mAh